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Space debris

Recommendations for Sustainable Orbital Debris Management

Conference room paper by the National Space Society

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Recommendations for Sustainable Orbital Debris Management

1. Executive Summary

1. Orbital debris refers to derelict human-made space objects that are still located in Earth's orbit. For the most part, orbital debris is only removed as a natural and unassisted process when its orbit decays enough for it to burn up in the Earth's atmosphere through aerothermal demise or for it to survive reentry and fall to the surface. For the latter, density, size, and materials with thermal stability, such as titanium, can all increase survivability. Atmospheric drag and solar radiation pressure are the main forces that cause the natural orbital decay. But at higher altitudes, orbital debris remains in Earth's orbits for much longer. At a perigee and apogee altitude of 20,000 kilometers, a debris object could take millions of years to decay. There are tens of thousands of softball-sized pieces, hundreds of thousands of marble-sized fragments, and over 100 million particles the size of a grain of salt currently in Earth's orbit.[1] Debris orbits the Earth at speeds of approximately 17,500 miles per hour, which is significantly faster than a bullet. *Id.* Even millimeter-sized particles can post a threat to space missions because they can't be tracked by ground-based sensors and spacecraft operators are thus unable to change course to avoid collisions with them. *Id.* Such a particle could crack windows, penetrate spacesuits, and erode surfaces. *Id.*

2. Orbital debris jeopardizes the safety of spacecraft and crew, the on-orbit operations of space systems, and the viability of human activity in Earth's orbit. Space-based applications are used for daily life and are critical to telecommunications (e.g., internet connectivity, cellular backhaul), emergency response services, weather forecasting, global-positioning systems, resource management for agriculture and gas, banking, etc. However, orbital debris has already reached critical mass. *Id.* As a result, collisional cascading will eventually happen even if no more objects are launched into Earth's orbit. *Id.* Each collision would generate additional orbital debris that creates a self-sustaining and increasing chain reaction of further collisions. *Id.* This could limit human access to space for hundreds of years. *Id.*

3. One of the National Space Society's ("NSS") main goals is to establish a stable orbital environment with safe and sustainable space operations in both the short-term and long-term. An impactful step towards this goal is creating a best practices regime that is widely implemented internationally to maintain a safe and stable orbital environment. Sustainable orbital debris management requires a comprehensive approach that will effectively use space situational awareness, space traffic management (e.g., norms of responsible behavior), mitigation, remediation, and on-orbit recycling. While active debris removal and in-space servicing, assembly, and manufacturing are in their technological infancy, future-proof policies can significantly support their development and eventual use.

4. Together, States can strengthen the international best practices regime by standardizing the application of debris mitigation guidelines, characterizing the legal and technical risks associated with active debris removal, and increasing thought leadership for on-orbit recycling. National debris mitigation guidelines should be transparent and effective for different spacecraft operators and on-orbit operations. Active debris removal should prioritize large rocket bodies in low Earth orbit with a clear State of Registry. The majority of orbital debris objects with the highest risk of causing major fragmentation events in low-Earth orbit are rocket bodies.[2] And on-orbit recycling needs to be more consistently and actively addressed in the international discourse surrounding orbital debris management.

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[1] National Aeronautics and Space Administration, *NASA's Efforts to Mitigate the Risks Posed by Orbital Debris*, Report No. IG-21-011 (2021).f

[2] Darren McKnight et al., *An Update of the Top 50 List*, ESA Space Debris Office 9-1 (2025).

2. Debris Mitigation

5. It's important to continue improving the best practices for mitigation since implementing the ideal regime requires an iterative and adaptive process. Mitigation already has the most developed international guidelines out of all the categories of orbital debris management. But mitigation is also likely the most impactful category to maintaining a safe and stable orbital environment. Moving forward it will be critical to increase transparency between the existing space regulatory regimes and then to create best practices that can be applied equally to all satellite operators.

6. First, States should publicly release not only the orbital debris mitigation guidelines they enforce, but also the standardized and verifiable methodologies that they require satellite operators to use to satisfy requirements for risk metrics with probability thresholds. Here, the term standardized means that the models and simulations used by satellite operators are widely accepted by the international space community. The term verifiable means that another State would be able to reproduce the results if (hypothetically) given the satellite operator's confidential assumptions. Similarly, the data inputs should reflect the real and current orbital environment. Risk metrics are highly sensitive to input parameters. Without standardized and verifiable methodologies, these metrics can be difficult to precisely compare across spacecraft from different satellite operators. These metrics could also inaccurately portray collision risk under the real-time orbital environment. For example, models make assumptions about the stochastic interactions of satellites and rely on the quality and frequency of the available orbital data, which creates opportunities for gamesmanship.

7. Second, best practices should take a performance-based approach that is technology-neutral. This includes defining the intended mitigation goal, while leaving flexibility for a satellite operator to choose how to achieve the goal in any given situation. Satellite operators would be able to leverage their bespoke technologies. They would be able to tailor the best practices to their specific mission profiles and operate under clear and predictable international industry conditions.

8. Third, best practices should assess collision risk for all satellite systems on a per-spacecraft basis, even when calculating probabilities for megaconstellations. Adopting collision risk metrics that are on a per-spacecraft basis better aligns incentives across all satellite operators. Each satellite, satellite constellation, and corresponding operator would be held to the same requirements for orbital debris mitigation. However, aggregate metrics result in uneven rules where safety requirements may vary depending on who operates an individual satellite. It is important to note that as the number of satellites increases, so does the accumulated collision risk.[3] Best practices should thus apply stricter collision risk thresholds for operators that have a large fleet of satellites (whether one constellation or multiple systems) on the per-spacecraft basis.

3. Debris Remediation

9. Remediation, and specifically active debris removal ("ADR"), is an important tool in orbital debris management. It will be necessary to reduce the likelihood of severe collisional cascading, especially as rare but inevitable major fragmentation events occur due to accidental satellite collisions and/or on-orbit anomalies. This decade alone, dozens of rocket bodies have been abandoned at end-of-life that will stay on orbit for more than 25 years.[4] Many of these post-mission disposals would now violate best practices codified in the Inter-Agency Space Debris Coordination Committee Space Debris Mitigation Guidelines ("IADC Guidelines").[5] As of 2021,

[3] National Aeronautics and Space Administration, *NASA Comments*, FCC IB Docket No. 18-313 (2020).

[4] *An Update of the Top 50 List.*

[5] Inter-Agency Space Debris Coordination Committee, *IADC Space Debris Mitigation Guidelines*, IADC-02-01 rev. 4 (2025).

the global compliance rate for post-mission disposal within 25-years average at only around 30 percent.[6] At the same time, the global cadence of rocket launches is drastically increasing. There is clearly a need for ADR to become a more reliable option in orbital debris management.

10. Increasing regulatory certainty while addressing operational risk will help strengthen the business case for both State and commercial operators so that they are incentivized to fully develop prospective ADR capabilities. One potential issue is that the jurisdiction and control clause of Article VIII of the Outer Space Treaty[7] extends state sovereignty to space objects indefinitely. This means that a space object is under the jurisdiction and control of a State of Registry even if its unclaimed or derelict on orbit at end-of-life. Most orbital debris is unclaimed and neither the owner, the operator, nor the State of Registry can be determined. Another more explicit issue is that ADR represents a high-risk mission profile. Rendezvous and proximity operations, especially for large, derelict rocket bodies, can generate significant debris in the event of an anomaly causing an accidental explosion or a collision. Such missions are also complicated and have a limited track record of success on orbit.

11. The United Nations Committee on the Peaceful Uses of Outer Space ("UN COPUOS") should establish a Debris Capture Index to increase regulatory certainty around which orbital debris objects have both the highest priority and the clearest legal mechanism for ADR. Orbital debris objects would be catalogued and ranked based on relevant factors, such as mission costs, mission risk (including the likelihood of causing a major fragmentation event), operational complexity, ADR opportunities for funding and/or revenue, orbital mass, tumbling properties, and the benefits of removal to the stability of the orbital environment. There are thousands of large orbital debris objects (e.g., rocket bodies) that are claimed by a State of Registry, registered on the UN Space Objects Registry, and well tracked by ground and/or in-situ sensors. The orbital debris objects at the top of the list would represent high positive impact, cost effective, and low risk options for initial missions.

12. The Debris Capture Index can also include information on whether the State of Registry for an orbital debris object has given express consent for a foreign remediation operator to conduct ADR. This is consistent with the customary international law principle of consent reflected in the International Law Commission's Articles on State Responsibility.[8] Article 20 states that "valid consent by a State to the commission of a given act by another State precludes the wrongfulness of that act in relation to the former State to the extent that the act remains within the limits of that consent." *Id.* In the future, the United Nations Office for Outer Space Affairs ("UN OOSA") Space Objects Registry[9] should include a new category covering whether a space object's State of Registry gives explicit consent for a foreign remediation operator to conduct ADR. This could apply when the spacecraft does not successfully complete post-mission disposal within 25 years of end-of-life. Finally, to address liability concerns a State of Registry could require the foreign remediation operator to cover it with sufficient third-party liability insurance.

13. States could also establish Protection and Indemnity Insurance Clubs ("P&I Clubs") to help address some of the operational risks associated with ADR. P&I Clubs, which are used in the maritime salvage regime, can spread and reduce risk for remediation operators.[10] P&I Clubs are non-profit mutual insurance associations that provide risk pooling for salvagers and other stakeholders. They are cheaper than traditional insurance and provide more coverage. *Id.* Members pay a sum of money

[6] *NASA's Efforts to Mitigate the Risks Posed by Orbital Debris.*

[7] United Nations, *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies*, 610 U.N.T.S. 205 (1967).

[8] International Law Commission, *Draft Articles on Responsibility of States for Internationally Wrongful Acts*, U.N. Doc. A/CN.4/SER.A/2001/Add.1 (2001).

[9] United Nations, *Convention on Registration of Objects Launched into Outer Space*, 1023 U.N.T.S. 15 (1975).

[10] National Space Society, *Space Debris Removal, Salvage, and Use: Maritime Lessons,* NSS Position Papers (2019).

into the club's pool. If there are major payouts, members will have to pay to replenish the pool. But if there are still funds in the pool at the end of the year, then each member will pay less the following year.

4. On-Orbit Recycling

14. Assuming the risk of accidental explosions can be appropriately reduced, on-orbit recycling is important to the long-term sustainability of the orbital environment. This is especially true as operators conduct more novel space activities and the number of on-orbit second stages and satellites drastically increases (e.g., megaconstellations). The future of in-space servicing, assembly, and manufacturing should be informed by the lessons learned from the global automobile industry where about 95% of today's retired vehicles enter the recycling process. An example of a circular space ecosystem is the emerging technologies that Astroscale, Nanoracks (i.e., Voyager Space), CisLunar Industries, and Neumann Space are collaboratively developing. NSS is optimistic that these companies will eventually be able to capture debris, cast it into metal feedstock rods, and then use that solid propellant in a Pulsed Cathodic Arc Thruster for electric propulsion.

15. Bringing on-orbit recycling to the forefront of UN COPUOS' discourse on orbital debris management could foster further research by non-governmental organizations, academia, and private startups on relevant emerging technologies and the corresponding cost-benefit analyses and use cases. The IADC Guidelines should highlight the future potential of on-orbit recycling to make space operations more sustainable. UN COPUOS should also expand on the first principle of the long-term sustainability of the orbital environment in the UN COPUOS Guidelines for the Long-term Sustainability of Outer Space Activities.[11]

5. Conclusion

16. The NSS urges States and non-governmental organizations to continue developing a best practices regime that is widely implemented internationally to maintain a safe and stable orbital environment. Sustainable orbital debris management requires a comprehensive approach that will effectively use space situational awareness, space traffic management (e.g., norms of responsible behavior), mitigation, remediation, and on-orbit recycling. Debris mitigation guidelines can further benefit from standardized, verifiable methodologies, technology-neutral requirements, and risk metrics with probability thresholds that are applied on a per-spacecraft basis. States can improve the business case for commercial active debris removal by creating public, continually updated Debris Capture Indices and establishing financial incentives and creative liability insurance opportunities. Finally, UN COPUOS and the IADC can support the long-term sustainability of the orbital environment by beginning to explicitly reference on-orbit recycling in their publications. NSS hopes that this next year brings the world a step closer to a stable, safe, and sustainable orbital environment.

The National Space Society ("NSS") is a US-based, international non-profit that is the preeminent citizen's voice on space exploration, development, and settlement. It publishes the award-winning Ad Astra magazine and contributes to thought leadership in the global space community. The issues surrounding orbital debris and its management have been a focus of its policy work.

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[11] United Nations, *Guidelines for the Long-term Sustainability of Outer Space Activities*, U.N. Doc. A/74/20, annex II (2019).